INTRODUCTION TO SYNTAX I

Session 3: Satisfying syntactic requirements via structure building

Timea Szarvas
University of Potsdam
timea.szarvas@uni-potsdam.de

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TODAY'S PLAN

- Combining elements into phrases
- Recursion
- Phrase structure
- Phrase types and terminology

STRUCTURE BUILDING

Predicates drive structure building by c-selecting arguments to satisfy their requirements.

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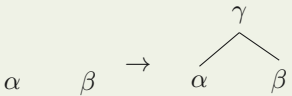
Structure-building operation Merge

Merge takes **exactly two** constituents α and β and merges them into a new constituent γ.



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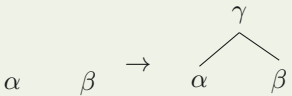

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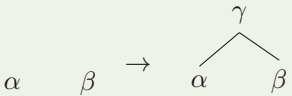

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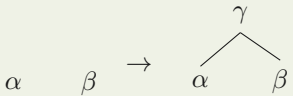

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Syntactic structures must not contain any [uX] features.

- How do the [uX] features of predicates 'disappear'?

(2) **Feature Checking under sisterhood**
A c-selection feature [uX] on a constituent Y is checked (and thus deleted) if Y is the sister of a constituent Z carrying a matching feature [X].

- Notation for checked features: ~~[uX]~~

FEATURE CHECKING: INTRANSITIVE VERBS

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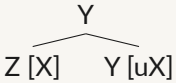

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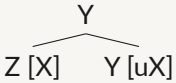

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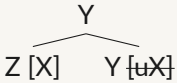

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  - **Full Interpretation:** [uN] on *arrive* must be checked (and deleted) by merging with a constituent carrying a [N] feature.

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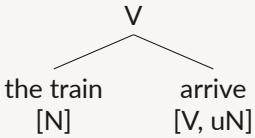

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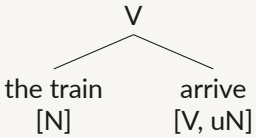

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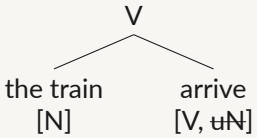

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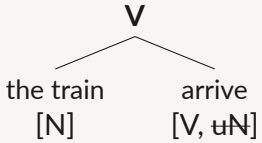

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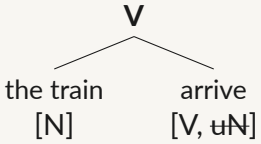

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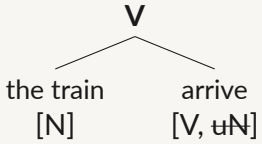

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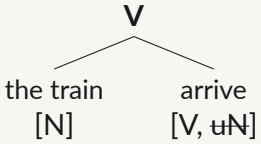

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- The daughter node from which the label inherited to the mother node is the **head** of the constituent.

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Inflectional features are inherited from the head

(12) a. My friend is cheerful.
b. My parents are cheerful.
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⇒ ✓*will* [V, uV]; ✗ *order* [V, uN] ⇒ *devour* [V] is the head!

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The element that c-selects projects!

⇒ Heads project their requirements, constraining what syntactic structures can be built.

FEATURE CHECKING: TRANSITIVE VERBS

What about heads with more than one c-selection feature?

- Let's see how *Hannah devoured the pizza* is derived:[1]

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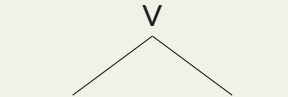

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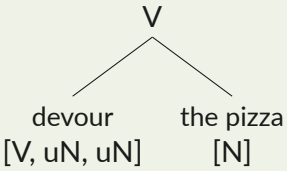

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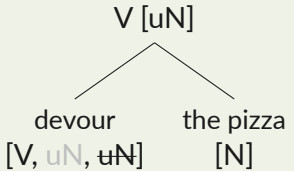

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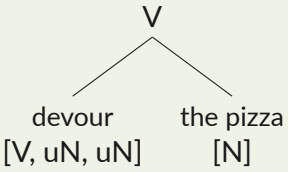

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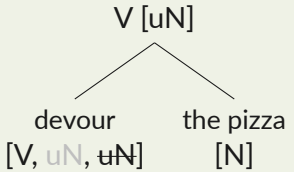

Step 1: *devour* [V, uN, uN] and *the pizza* [N] are inserted from the numeration.
Step 2: Merge combines *devour* with *the pizza*.
Step 3: one c-selectional feature on *devour* [V, uN, uN] is checked+deleted; the other one is inherited to the mother V.

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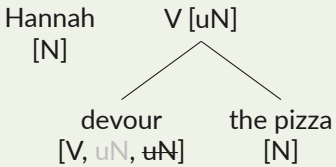

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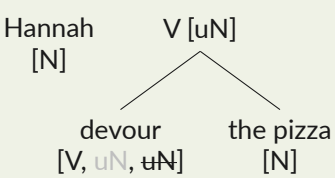

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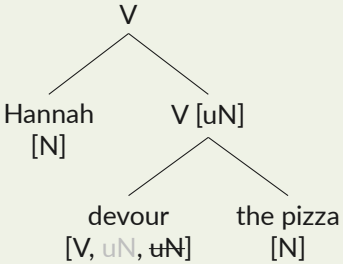

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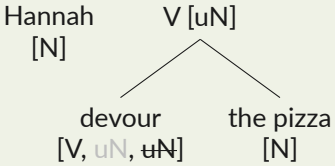

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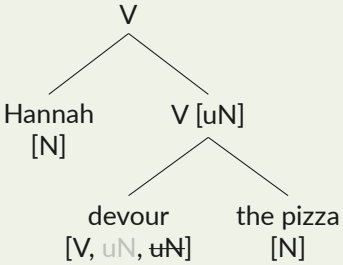

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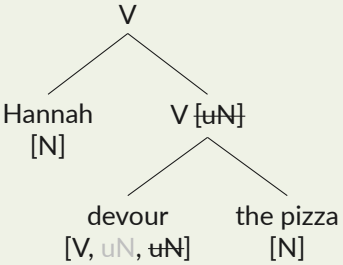

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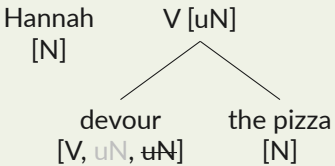

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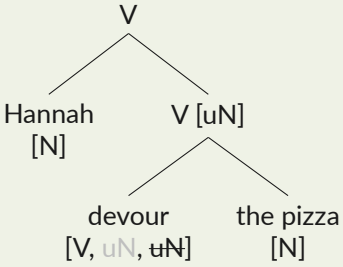

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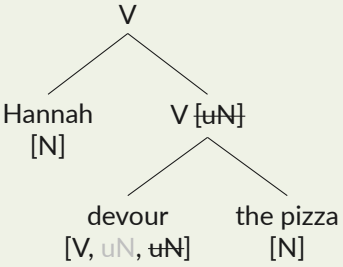

Step 4: *Hannah* is inserted from the numeration.
Step 5: *Hannah* is merged with *devour the pizza* [uN].
Step 6: the c-selectional feature on *devour the pizza* [uN] is checked+deleted ⇒ the derivation is complete.

Recall session 1:

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- This is precisely what we did when we derived *Hannah devoured the pizza* $\Rightarrow$ [Hannah + [devour + the pizza]].

RECURSION

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Formalization via Merge:

⇒ The complexity of syntactic structures results from repeated applications of Merge.

⇒ The constituents' selectional criteria constrain what they can Merge with.

NON-VERBAL HEADS

Two concepts not to be confused!

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⇒ **Valency** independently requires predicates to combine with a specific number of arguments, to ensure that the expressed event is **semantically meaningful**.

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⇒ **Valency** independently requires predicates to combine with a specific number of arguments, to ensure that the expressed event is **semantically meaningful**.

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- In constituents like *the parents of my friend* (12) and *owners of a pig* (13), the heads *the parents* and *owners* select a preposition, but they could also appear without it.

(22) a. The parents are responsible for the child.
b. Owners are free to choose the tenants they like.

EXERCISE (10 MIN)

(23) the books on the table

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   - What is the head?
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2. Individually:
   - What is the structure of the whole constituent? Draw a tree introducing each element step-by-step
   - Once you have the tree: label all of the nodes based on category and selectional features.
   - **Hint 1:** no need to overthink the internal structure of *the books* or *the table*.
   - **Hint 2:** if you have trouble figuring out which elements belong 'closer' together, try the constituency tests from session 1!

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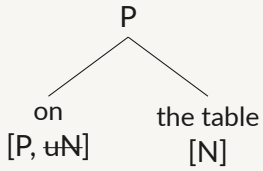

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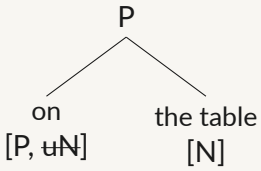

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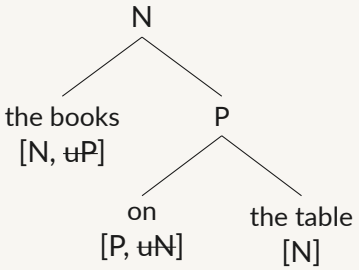

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Next, we will learn more about 'projection'.

- We have seen that **remaining c-selectional features project**:

⇒ In (18), the remaining [uN] projected from *devour* to V.

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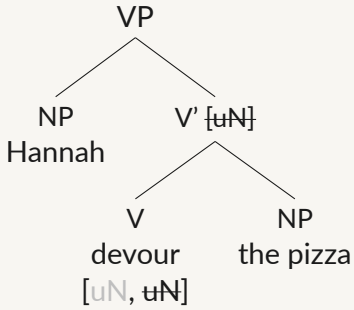

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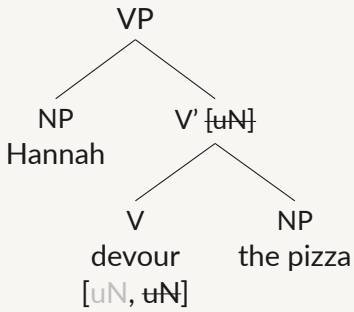

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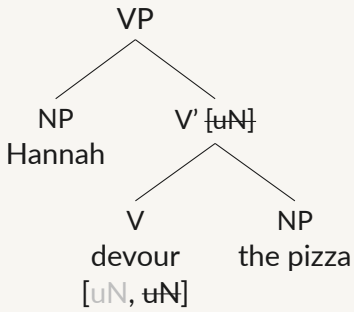

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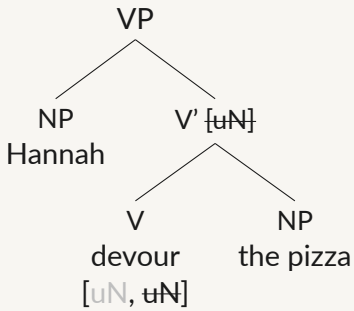

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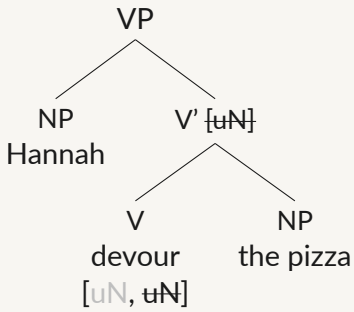

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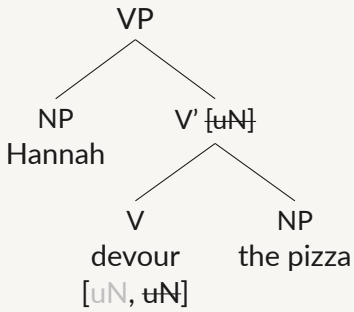

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- *VP (Hannah devour the pizza):* maximal projection

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- Implicitly, I showed you that **the smallest possible constituents combine first**:

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- **Complements are maximal projections** (*the pizza* has no [uX]] features) and **sisters of minimal projections** (*devour* is a minimal projection).

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A note on linearization

- The left-to-right order in which our constituents appeared in the hierarchical structures we saw so far happened to correspond to the order in which they linearly appear in the sentence.

⇒ This is not always the case!

LINEARIZATION

(28) $[_{\gamma}\ \alpha\ \beta]$

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Merge is insensitive to linear order.

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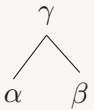

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$\Rightarrow$ **A specifier cannot intrude between a head and its complement!**

Ask away!